

Both $p < 1$ Atomic-Block Questions Fail on an Atomic Filtration

Solution packet for arXiv:1407.5451, Problem 3.1

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Status. Counterexample likely valid, subject to human review. For every $0 < p < 1$ and $1 < q < \infty$, both assertions in the source problem fail on the same countably atomic probability space with one fixed filtration.

1 Source problem

Conde-Alonso and Parcet define (p, q) -atomic blocks for $0 < p < 1$, then ask in Problem 3.1 whether

$$H_{\text{atb},q}^p(\Omega) = H_p(\Omega) \quad \text{and} \quad H_p(\Omega)^* = \Lambda_{p,q}(\Omega).$$

Here H_p is defined by the martingale square function. The proposed Lipschitz norm is

$$\|h\|_{\Lambda_{p,q}} = \sup_{k \geq 1, A \in \Sigma_k} \frac{1}{\mu(A)^{1/p-1}} \left[\left(\frac{1}{\mu(A)} \int_A |h - \mathbb{E}_k h|^q d\mu \right)^{1/q} + \|dh_k\|_\infty \right]. \quad (1)$$

The question and the definitions immediately preceding it occur on page 17 of the source paper [1].

For a (p, q) -subatom supported on $A_j \in \Sigma_{k_j}$, the source requires

$$\|a_j\|_q \leq \mu(A_j)^{1-1/p-1/q'} \frac{1}{k_j - k + 1} = \mu(A_j)^{1/q-1/p} \frac{1}{k_j - k + 1}. \quad (2)$$

The size of one block $b = \sum_j \lambda_j a_j$ is the infimum of $\sum_j |\lambda_j|$, while the outer decomposition of a function into blocks is aggregated in ℓ_p .

2 Proof intuition

The definition has an exponent mismatch in how it groups pieces. Inside one block, arbitrarily many subatoms are charged in ℓ_1 . For $p < 1$, a martingale square function can see the same pieces in ℓ_p , which is much larger.

The concrete gadget has two signs. At level two it equals $+1$ or -1 . At level three it usually cancels back to zero, but with probability ε it jumps to size $1/\varepsilon$ in order to preserve the martingale condition. Its terminal L_p norm tends to zero like $\varepsilon^{1/p-1}$, and the proposed atomic-block norm has the same small upper bound. Its square function, however, remembers the level-two jump of size one on the whole block. Countably many disjoint copies give a function in $L_p \cap H_{\text{atb},q}^p$ but not in H_p .

For the dual assertion, use a decaying sum of the level signs. Its decay is chosen so that testing every martingale difference costs at most its local L_p square-function mass. Formula (1) nevertheless assigns it infinite norm: at late levels its nonzero global jump is multiplied by a negative power of the measure of an already resolved, extremely small atom.

3 The counterexample

Theorem 1. Fix $0 < p < 1$ and $1 < q < \infty$. There is a countably atomic probability space with a fixed filtration for which

- (i) $H_{\text{atb},q}^p(\Omega) \neq H_p(\Omega)$, even as sets;
- (ii) $H_p(\Omega)^* \neq \Lambda_{p,q}(\Omega)$, with $\Lambda_{p,q}$ defined by (1).

Proof. Put

$$w_n = 2^{-n}, \quad \varepsilon_n = 2^{-n/(1-p)} \quad (n \geq 1).$$

Thus $\sum_n w_n = 1$, $0 < \varepsilon_n \leq 1/2$, and

$$\varepsilon_n^{1-p} = 2^{-n}. \quad (3)$$

Partition Ω into atoms $A_n^+, C_n^+, A_n^-, C_n^-$, with

$$\mu(A_n^\pm) = \frac{w_n \varepsilon_n}{2}, \quad \mu(C_n^\pm) = \frac{w_n(1 - \varepsilon_n)}{2}.$$

Let $B_n^\pm = A_n^\pm \cup C_n^\pm$, put $B_n = B_n^+ \cup B_n^-$, and let $R_n = \bigcup_{j \geq n} B_j$. Define an increasing filtration by declaring that the atoms of Σ_{2n-1} are

$$A_j^\pm, C_j^\pm \quad (j < n), \quad R_n,$$

and that the atoms of Σ_{2n} are

$$A_j^\pm, C_j^\pm \quad (j < n), \quad B_n^+, B_n^-, R_{n+1}.$$

Thus Σ_1 is trivial. From level $2n-1$ to level $2n$ the filtration reveals the sign in the n -th block, and at level $2n+1$ it reveals whether the rare event occurred.

Define

$$g_n = \varepsilon_n^{-1}(\mathbf{1}_{A_n^+} - \mathbf{1}_{A_n^-}). \quad (4)$$

The two signs have equal mass, so $\mathbf{E}_{2n-1}g_n = 0$, while

$$\mathbf{E}_{2n}g_n = \mathbf{1}_{B_n^+} - \mathbf{1}_{B_n^-} =: r_n, \quad \mathbf{E}_{2n+1}g_n = g_n.$$

Consequently the only nonzero differences are $dg_{n,2n} = r_n$ and $dg_{n,2n+1} = g_n - r_n$. The square function $S(g_n)$ is $\sqrt{2}$ on $C_n^+ \cup C_n^-$, and it is

$$\left(1 + \left(\frac{1 - \varepsilon_n}{\varepsilon_n}\right)^2\right)^{1/2}$$

on $A_n^+ \cup A_n^-$. Therefore

$$\|g_n\|_p^p = w_n \varepsilon_n^{1-p}, \quad (5)$$

$$\|g_n\|_{H_p}^p = w_n \left[(1 - \varepsilon_n)2^{p/2} + \varepsilon_n \left(1 + \left(\frac{1 - \varepsilon_n}{\varepsilon_n}\right)^2\right)^{p/2} \right] \geq 2^{p/2-1} w_n. \quad (6)$$

We next verify that g_n is a very cheap (p, q) -atomic block. Set $m_n = w_n \varepsilon_n/2$ and

$$a_n^+ = \frac{m_n^{-1/p}}{3} \mathbf{1}_{A_n^+}, \quad a_n^- = -\frac{m_n^{-1/p}}{3} \mathbf{1}_{A_n^-}, \quad \lambda_n = \frac{3m_n^{1/p}}{\varepsilon_n}.$$

Both subatoms are supported at level $k_j = 2n + 1$, while the block has cancellation at level $k = 2n - 1$. Moreover

$$\|a_n^\pm\|_q = \frac{1}{3} m_n^{1/q-1/p} = m_n^{1/q-1/p} \frac{1}{(2n+1) - (2n-1) + 1},$$

so (2) holds with equality, for every $q > 1$. Since $g_n = \lambda_n a_n^+ + \lambda_n a_n^-$,

$$|g_n|_{\text{atb},q}^p \leq 2\lambda_n = 6 \cdot 2^{-1/p} w_n^{1/p} \varepsilon_n^{1/p-1}. \quad (7)$$

Let $t_n = w_n^{-1/p}$. The finite martingales

$$\begin{aligned} F_{2N-1} &= \sum_{n < N} t_n g_n, \\ F_{2N} &= \sum_{n < N} t_n g_n + t_N r_N, \\ F_{2N+1} &= \sum_{n \leq N} t_n g_n \end{aligned}$$

are integrable and compatible with the filtration. They converge pointwise and in L_p to

$$f = \sum_{n \geq 1} t_n g_n.$$

By (5) and disjointness,

$$\|f\|_p^p = \sum_n t_n^p w_n \varepsilon_n^{1-p} = \sum_n 2^{-n} < \infty.$$

Using the blocks $t_n g_n$, equations (7) and (3) give

$$\|f\|_{H_{\text{atb},q}^p}^p \leq (6 \cdot 2^{-1/p})^p \sum_n \varepsilon_n^{1-p} < \infty.$$

Thus the series is an admissible atomic-block decomposition of the L_p -function f . On the other hand, square functions are supported on the disjoint sets $B_n = B_n^+ \cup B_n^-$. Hence (6) implies

$$\|f\|_{H_p}^p = \sum_n t_n^p \|g_n\|_{H_p}^p \geq 2^{p/2-1} \sum_n t_n^p w_n = 2^{p/2-1} \sum_n 1 = \infty.$$

This proves (i).

For (ii), put $\alpha = 1/p - 1 > 0$, $\beta_n = (w_n/2)^\alpha$, and take the bounded function

$$\phi = \sum_{n \geq 1} \beta_n r_n.$$

Its only nonzero martingale differences are $d\phi_{2n} = \beta_n r_n$. On finite martingales define

$$L_\phi(h) = \sum_{n \geq 1} \beta_n \int (dh_{2n}) r_n d\mu;$$

this agrees with $\int h \phi d\mu$ whenever the latter is directly defined. Write $c_{n,\pm}$ for the two constant values of dh_{2n} on $B_n^\pm$, and put $m'_n = w_n/2$. Since $\beta_n = (m'_n)^{1/p-1}$,

$$\begin{aligned} \left| \beta_n \int (dh_{2n}) r_n d\mu \right| &\leq \beta_n m'_n (|c_{n,+}| + |c_{n,-}|) \\ &\leq \left(\int_{B_n} |dh_{2n}|^p d\mu \right)^{1/p} =: x_n. \end{aligned}$$

For $p < 1$, $\sum_n x_n \leq (\sum_n x_n^p)^{1/p}$. Therefore

$$|L_\phi(h)| \leq \left(\sum_n \int_{B_n} |dh_{2n}|^p d\mu \right)^{1/p} \leq \|h\|_{H_p}.$$

Thus L_ϕ extends continuously and ϕ represents an element of H_p^* .

But $\phi \notin \Lambda_{p,q}$. In (1), take $k = 2n$ and the already resolved atom $A = A_{n-1}^+$, for $n \geq 2$. Since $\|d\phi_{2n}\|_\infty = \beta_n$, the corresponding expression is at least

$$\frac{\beta_n}{(w_{n-1}\varepsilon_{n-1}/2)^\alpha} = \left(\frac{1}{2\varepsilon_{n-1}} \right)^\alpha \rightarrow \infty.$$

This proves (ii). □

4 Verification notes

The proof uses only the definitions printed in the source. In particular:

- the filtration is fixed once and for all; every finite stage of the separating martingale is integrable, and the finite stages converge in L_p to the displayed terminal function;
- every subatom satisfies the source normalization with equality;
- the separating function f belongs to L_p , so the counterexample respects the set-level requirement in the definition of $H_{\text{atb},q}^p$;
- the same construction works for every finite $q > 1$, because the q -dependence cancels in the indicator normalization;
- the dual counterexample uses exactly the global jump term as it appears inside the brackets in the source’s displayed $\Lambda_{p,q}$ norm.

The accompanying script `code/verify_counterexample.py` evaluates the finite-block formulas and partial sums for several choices of p, q . It is a sanity check, not a substitute for the proof.

5 Novelty and limitations

A bounded search was performed in the run indexes and on arXiv using 1407.5451, the exact notation $H_{\text{atb},q}^p$, the paper title, and the phrases “ (p, q) -atomic blocks” and “atomic block H_p martingale”. No later paper settling Problem 3.1 was found. The novelty claim is therefore bounded rather than exhaustive.

The theorem answers Problem 3.1 negatively for arbitrary filtrations. It does not rule out a repaired theorem under additional regularity, a restriction on how subatoms may be grouped, or a modified Lipschitz norm in which the jump term is not multiplied by the measure of an unrelated set.

Human-review recommendation. Verify the source transcription of the subatom exponent and the placement of $\|df_k\|_\infty$ in (1); then check the two conditional expectations in (4). All remaining estimates are exact scalar calculations.

References

- [1] J. M. Conde-Alonso and J. Parcet, *Atomic blocks for noncommutative martingales*, arXiv:1407.5451 (2014), Problem 3.1, p. 17.

$$\gtrsim \frac{1}{\mu(A)} \left(\frac{1}{2} - \frac{1}{32} - \frac{1}{16} \right) \|f\|_{\text{BMO}}^\alpha \mu(A) \gtrsim \|f\|_{\text{BMO}}^\alpha.$$

This is the last possible case and completes the proof of Theorem A for $p = \infty$. $\square$

3. Open problems

When $0 < p < 1$, one can extend the definition of atomic blocks to (p, q) -atomic blocks. Given $1 < q < \infty$, b is called a (p, q) -atomic block when b is Σ_1 -measurable or there exists $k \geq 1$ such that the following properties hold:

- $\mathbf{E}_k(b) = 0$,
- $b = \sum_j \lambda_j a_j$ where
 - $\text{supp}(a_j) \subset A_j$,
 - $\|a_j\|_q \leq \mu(A_j)^{1-\frac{1}{p}-\frac{1}{q}} \frac{1}{k_j-k+1}$,
 for certain $k_j \geq k$ and $A_j \in \Sigma_{k_j}$.

As in the case of $p = 1$, set $|b|_{\text{atb},q}^p = \|b\|_p^p$ if $b \in L_p(\Omega, \Sigma_1, \mu)$ and

$$|b|_{\text{atb},q}^p = \inf_{\substack{b = \sum_j \lambda_j a_j \\ a_j \text{ } (p,q)\text{-subatom}}} \sum_{j \geq 1} |\lambda_j|$$

otherwise. Finally, we define

$$\mathbf{H}_{\text{atb},q}^p(\Omega) = \left\{ f \in L_p(\Omega) \mid f = \sum_i b_i, \text{ } b_i \text{ martingale } (p, q)\text{-atomic block} \right\},$$

equipped with the following quasi-norm

$$\|f\|_{\mathbf{H}_{\text{atb},q}^p(\Omega)} = \inf_{\substack{f = \sum_i b_i \\ b_i \text{ } (p,q)\text{-atomic block}}} \left[\sum_{i \geq 1} (|b_i|_{\text{atb},q}^p)^p \right]^{\frac{1}{p}}.$$

The spaces $\mathbf{H}_{\text{atb},q}^p(\Omega)$ defined above are quasi-Banach subspaces of $L_p(\Omega)$. One can follow almost *verbatim* the steps in the proof of Theorem A to conclude that the set of linear continuous functionals acting on $\mathbf{H}_{\text{atb},q}^p(\Omega)$ can be identified with the Lipschitz type class $\Lambda_{p,q}(\Omega)$ of functions with finite norm

$$\|f\|_{\Lambda_{p,q}(\Omega)} = \sup_{\substack{k \geq 1 \\ A \in \Sigma_k}} \frac{1}{\mu(A)^{\frac{1}{p}-1}} \left[\left(\int_A |f - \mathbf{E}_k f|^q d\mu \right)^{\frac{1}{q}} + \|df_k\|_\infty \right].$$

Notice that when $p \rightarrow 1$, the norm in $\Lambda_{p,q}(\Omega)$ tends to the norm in $\text{BMO}(\Omega)$. This motivates our first problem, which is somehow analogous (in the context of atomic blocks of this paper) to Problem 3 in [1].

Problem 3.1. *Do we have*

$$\mathbf{H}_{\text{atb},q}^p(\Omega) = \mathbf{H}_p(\Omega)$$

for $0 < p < 1$ and $q > 1$? Moreover, do we have $\mathbf{H}_p(\Omega)^* = \Lambda_{p,q}(\Omega)$?

Our main result shows that a function in $\mathbf{H}_1(\Omega)$ can be decomposed into atomic blocks similar to the ones appearing in the definition of $\mathbf{H}_{\text{atb}}^1(\mathbb{R}^n, \mu)$, the atomic block Hardy space of Tolsa [26]. In the proof given in Section 1, we make use of Davis decomposition for martingales. It is natural to ask whether we can find a

Figure 1: The definition and Problem 3.1 on page 17 of arXiv:1407.5451.